# 6.7 Modeling with Exponential and Logarithmic Functions

Name _____ Period ___ Date _____

Algebra II 6.7 · Student sheet

**I can…**

- tell whether a table shows a linear function, an exponential function, or neither.
- write an exponential function through two points.
- use technology to find an exponential model and make predictions.

**Vocabulary** · fill in the missing word

**Constant differences** — equal _____ and equal differences in y: linear

**Constant ratios** — equal x-steps and equal ratios of y: _____

**Exponential regression** — a _____ or Desmos model $y = ab^x$ that best fits data

## Part 1 · Linear or exponential?

For each table (x = 0, 1, 2, 3, 4), find the differences and the ratios of consecutive y-values.

| y-values | Differences | Ratios | Linear, exponential, or neither? |
|---|---|---|---|
| 2, 5, 8, 11, 14 | | | |
| 4, 12, 36, 108, 324 | | | |
| 1, 3, 7, 13, 21 | | | |

## Part 2 · Notes: fill in the blanks

Constant _____ → linear. Constant _____ → exponential.

To find $y = ab^x$ through two points: substitute both points, then _____ one equation by the other to find b.

In Desmos, exponential regression is typed as _____.

## Part 3 · Examples (work them with the class)

### Example 1

Is the table linear, exponential, or neither?

x: 0 1 2 3 4

y: 5 10 20 40 80

### Example 2

Is the table linear, exponential, or neither?

| x: | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| y: | 3 | 5 | 9 | 15 | 23 |

### Example 3

Write an exponential function $y = ab^x$ whose graph passes through (1, 6) and (3, 54).

### Example 4

Write an exponential function $y = ab^x$ whose graph passes through (2, 12) and (4, 3).

### Example 5

A town's population (in hundreds) t years after 2020 is shown. Use exponential regression to find a model. Predict the population in 2026.

| t: | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| y: | 120 | 150 | 185 | 232 | 290 |

## Part 4 · Partner practice

### Partner 1

Is the table linear, exponential, or neither? If exponential, write the function.

| x: | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| y: | 2 | 6 | 18 | 54 |

### Partner 2

Write an exponential function through (0, 4) and (2, 36).